Year 12 Maths - Binomial expansion

Similar questions - Terms and coefficients

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 In this question k is a non-zero constant.

(a) Find and simplify the first three terms, in ascending powers of x, of the expansion of $(3+kx)^5$. **[3]**

(b) Given that the coefficient of x^2 is three times the coefficient of x, find the value of k. **[3]**

2

(a) Find the first four terms, in ascending powers of x, of the expansion of $(1+3x)^5$. **[3]**

(b) Hence find the coefficient of x^3 in the expansion of $(2-x)(1+3x)^5$. **[3]**

Question 1

(a) $3^5 + \binom{5}{1}3^4kx + \binom{5}{2}3^3k^2x^2 = 243 + 405kx + 270k^2x^2$

(b) $270k^2 = 3 \times 405k \Rightarrow 270k = 1215 \Rightarrow k = \dfrac{9}{2}$

Question 2

(a) $1 + 5(3x) + 10(3x)^2 + 10(3x)^3 = 1 + 15x + 90x^2 + 270x^3$

(b) Coefficient of x^3: $2 \times 270 + (-1) \times 90 = 540 - 90 = 450$